

**CSC331: Data Structures – BMCC Fall 2025**  
**Professor Byron**  
**Program #4 specifications: Priority queue**  
**Due: 1159pm Friday 11-21-2025**  
**No credit for late submission**

Your task for this assignment is to develop a ticket scheduling application. The application will use an array-based heap to implement a priority queue data structure in C++. This will be accomplished utilizing a user-defined Heap class.

1. Implement a transaction-based priority queue data structure using C++. The program will be interactive. A prompt will be displayed requesting each input transaction. Each input transaction will be entered at the command line and results will be displayed on the console.
2. Each input transaction will represent either (1) a new ticket request transaction or (2) a service transaction. A new ticket request transaction will be represented by a string consisting of a string name and an integer age separated by a comma. Each new ticket request is scheduled based on priority in first-come-first-serve order. The highest priority is given to children (age less than 16), followed by seniors (age greater than 64) and then others (age greater than 15 and less than 65). Each new ticket is placed in a priority queue and will be serviced by the next available service agent. A new ticket request received when the queue is full (10 ticket maximum) results in an error message "error: full" displayed, otherwise, the message "queued" is displayed.

A service transaction results in a ticket being removed from the priority queue and the name from the transaction displayed on the console. A service transaction received when the queue is empty results in an error message "error: empty" displayed.

A transaction containing the string "end-of-file" will signal the end of the input. Your program will display the number of tickets remaining in the queue when the "end-of-file" transaction is processed.

Sample input transactions and output expected using a priority queue:

|           |                        |
|-----------|------------------------|
| service   | (output: error: empty) |
| Brian,79  | (output: queued)       |
| Amina,39  | (output: queued)       |
| Oumy,9    | (output: queued)       |
| service   | (output: Oumy)         |
| Rafiul,69 | (output: queued)       |
| Bilal,49  | (output: queued)       |
| Simon,19  | (output: queued)       |
| service   | (output: Brian)        |

|              |                               |
|--------------|-------------------------------|
| service      | (output: Rafiul)              |
| service      | (output: Amina)               |
| service      | (output: Bilal)               |
| Cassandra,89 | (output: queued)              |
| Santiago,10  | (output: queued)              |
| Kimberly,30  | (output: queued)              |
| service      | (output: Santiago)            |
| end-of-file  | (output: 3 tickets remaining) |

3. The program will be run at the command prompt by navigating to the directory containing the executable version of the program after the program is compiled. Each input transaction will be correctly formatted. Use the heap class heap331\_f25.h header file located in Brightspace. Use “include” in your program to retrieve the header file. No changes are permitted in the header file.
4. Your C++ program file should be named csc331\_section\_prog4\_lastname.cpp. Your program should contain comments starting on line 1 of the program containing the following information:
  - a. course ID and section
  - b. your full name
  - c. the program file name
  - d. the program assignment number and due date
  - e. the program purpose

You are encouraged to add additional comments throughout the program that you feel might be helpful to the reader of your source code. Your program code indentation must be neat for readability.

5. Submit your C++ program (cpp file) as an unzipped attachment to an email message to kbyron@bmcc.cuny.edu using a subject in this form: “csc331\_section\_prog4\_lastname”. A link to a file located on a remote server will not be accepted. If you submit your assignment at least one week early, it will be graded. If you wish, you may then re-submit your complete assignment prior to the due date for a potentially higher score.
6. Do your own work. Students submitting copies of the same program will receive grades of zero for the assignment. Do not share your code until you have received your final grade for the course at the end of the semester.